

EMILIO RANUCOLI

Machine Learning Engineer | Python Developer | Data & AI Specialist

Barcelona, Spain | Italian Citizen - EU Work Authorization (no sponsorship required)

Open to Remote EU & Hybrid | Trilingual: English (Fluent), Spanish (Native), Italian (Native)

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PROFESSIONAL SUMMARY

Machine Learning Engineer and Python Developer based in Barcelona, Spain, with hands-on experience building production ML systems, ETL data pipelines and microservices for Booking.com, Accenture and enterprise clients across the EU. Specialized in Python, Scikit-learn, XGBoost, TensorFlow, FastAPI and AWS, with a strong record of measurable impact (12% lift in promotional targeting accuracy at Booking.com, 15% supply-chain cost reduction at Accenture, 2M+ daily transactions processed via near-real-time ETL). Expertise spans end-to-end MLOps pipeline construction, scalable AI/LLM integrations and robust system architecture. Founder of Ranuk IT Solutions, delivering custom software and ML products to clients in 4 countries. Italian citizen with full EU work authorization (no sponsorship required).

"Combines technical rigor with an uncommon ability to understand the problem before writing code. Recommended 100%." - Delfina Bugliotti, Product Manager @ Booking.com

TECHNICAL SKILLS

Languages & ML: Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras, XGBoost, PyTorch), SQL, JavaScript / TypeScript, C#

Machine Learning: Supervised & Unsupervised Learning, Deep Learning, Neural Networks, NLP, Sentiment Analysis, Time Series Forecasting, MLOps

Data Engineering: ETL Pipelines, Data Wrangling, Data Modeling, Data Warehouse, RESTful APIs, JSON, Kafka, Streaming Data

Backend & Web: FastAPI, Flask, Node.js, React, Tailwind CSS, HTML5 / CSS3, Microservices, GraphQL

BI & Visualization: Power BI (PL-300 Certified), DAX, Tableau, Looker, Grafana, Streamlit, Plotly

Databases: PostgreSQL, MongoDB, MySQL, SQL Server, SAP

Cloud & DevOps: AWS (Cloud Practitioner Certified), Microsoft Azure (AI Fundamentals), GCP, Docker, Kubernetes, Git / GitHub, CI/CD

Testing & Methodologies: JUnit, TDD, REST, SOAP, Agile / Scrum, Code Reviews

PROFESSIONAL EXPERIENCE

Machine Learning Engineer (Contract) | Booking.com

Amsterdam, Netherlands · May 2025 - Oct 2025

- Led the development of a dynamic pricing optimization system, deploying Machine Learning models (Python, Scikit-learn, XGBoost, neural networks) for real-time price adjustments based on demand, competition and customer behavior.
- Analyzed 5M+ historical booking records to identify patterns that improved customer segmentation and promotional campaign effectiveness across multiple markets.
- Integrated production ML models into Booking.com's AWS environment with continuous monitoring, ensuring scalability and reproducibility of predictions.
- Built a GraphQL backend layer to reduce API latency and serve cached features to downstream services.
- Contributed to a 12% improvement in promotional targeting accuracy by mastering internal forecasting tools and tuning model features.

Founder & Lead Engineer | Ranuk IT Solutions

Barcelona, Spain (Remote) · 2024 - Present

- Founded and operate a tech consultancy delivering custom software and applied Machine Learning solutions to clients in Argentina, Spain, Italy, and the United States.
- Delivered 28+ in-house projects across web, backend, automation, and ML, owning the full lifecycle from product discovery to deployment and continuous support.
- Eurobrico S.p.A. (Major Italian DIY Retail Chain / External Consultancy, Nov 2024 - Apr 2025): Deployed Python inventory management applications and data integration workflows, automating retail operations and reducing manual errors; executed a complete UX/UI web redesign and front-end prototype to modernize the customer journey.
- NOT A ROBOT (Audiovisual Production): Designed and built the full web platform powering their brand presence and client showcase.
- Bahay Design (Architecture Studio, Italy): Delivered a high-end responsive website with same-day iteration cycles requested directly by the client.
- GARYCIO: Built a production-ready backend service with WhatsApp bot integration, PostgreSQL persistence, and automated PDF report generation.
- ranuk.dev (Personal Engineering Brand): Features an interactive, in-browser Neural Network Lab with configurable learning rates, epochs, and hidden neurons showing real-time decision-boundary visualization.
- Core stack: TypeScript, React, Node.js, Python, PostgreSQL, Scikit-learn, XGBoost, TensorFlow, PyTorch, MLOps.

Python Developer - Supply Chain Integration (Main role) | Accenture

Rome, Italy (On-site) · Nov 2024 - Apr 2025

- Designed and built Python microservices to integrate supply-chain data between SAP, Oracle and legacy ERPs for a major logistics client.
- Engineered automated ETL pipelines processing 2M+ daily transactions, reducing data latency from 24 hours to near real-time.
- Developed REST APIs for inventory synchronization across warehouses, achieving 99.7% data consistency across 50+ locations.
- Implemented Grafana dashboards and alerting systems for pipeline health monitoring and SLA compliance tracking.
- Stack: Python, FastAPI, SAP, Oracle, PostgreSQL, Docker, Grafana, REST APIs.

Data Science Analyst / Sr. Analyst - Supply Chain & Operations | Accenture

Argentina (Remote) · Jun 2024 - Nov 2024

- Developed predictive and optimization models for supply-chain operations using Python, SQL and Power BI.
- Implemented Machine Learning models for demand forecasting and inventory optimization, achieving a 15% operational cost reduction.
- Built interactive dashboards in Tableau and Power BI to visualize critical KPIs and support data-driven decision-making.
- Worked with multidisciplinary global teams under Agile / Scrum methodologies.

Data Analyst & Power BI Specialist | Government of Córdoba

Córdoba, Argentina · Nov 2023 - Jun 2024

- Designed and maintained 15+ Power BI dashboards tracking KPIs for public works and infrastructure programs.
- Performed data cleaning and modeling on 200K+ record datasets, improving report accuracy by 30%.
- Automated certification reporting workflows, reducing manual effort by 40% and freeing analyst time for higher-value work.

Purchasing & Inventory Analyst | Farmasut

Córdoba, Argentina · Nov 2015 - Oct 2023

- Conducted inventory audits and sales analysis with Tableau, reducing expiration losses by 15%.
- Optimized delivery routes through data analysis, decreasing logistics costs and improving fulfillment SLAs.

KEY PROJECTS

GARYCIO - WhatsApp Backend Bot | github.com/RanuK12

Stack: TypeScript, Node.js, WhatsApp API, PostgreSQL, PDF Generation

Production-grade backend service with WhatsApp integration and automated PDF reporting deployed for an international enterprise client.

Crypto Analysis Dashboard | github.com/RanuK12

Stack: Python, Streamlit, Plotly, Binance API, Machine Learning, Telegram Bot

Cryptocurrency market analysis platform with real-time data ingestion, algorithmic trading strategy backtesting and a modern interactive web interface.

JobConnect - AI Job Finder | github.com/RanuK12

Stack: Flask, NLP, Web Scraping, Tailwind CSS

Web platform that uses NLP to match candidate CVs with job openings across multiple platforms, ranking results by semantic fit.

EDUCATION

Systems Engineering - Universidad Tecnológica Nacional (UTN), Argentina

CERTIFICATIONS

- Microsoft Power BI Data Analyst (PL-300) — Microsoft (2024)
- AWS Cloud Practitioner — Amazon Web Services (2024)
- Microsoft Azure AI Fundamentals (AI-900) — Microsoft (2024)
- Machine Learning with Python — IBM (2025)
- Deep Learning Fundamentals — IBM (2025)
- Deep Learning with TensorFlow — IBM (2025)
- Intermediate Machine Learning — Kaggle (2024)

LANGUAGES

English: Fluent (Professional Working Proficiency) **Spanish:** Native **Italian:** Native

PUBLICATIONS

"Y así voy tejiendo mi camino" - Memoir, self-published 2024. Available on Kindle.